

Masoud SADRNEZHAAD

PhD Student at Linköping University

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I am a PhD candidate in the Complex Software-Intensive Systems Research Group at Linköping University. My research focuses on the intersection of cyber-physical systems modeling and LLMs. Before my PhD, over a 12-year span, I held roles as a web developer, technical lead, software architect, recommender systems researcher, programming instructor, and Linux server administrator.

Competencies: Software Engineering Agentic AI Recommender Systems Modelica Web Development Linux

EDUCATION

- Oct 2024 Now** **Doctor of Philosophy - PhD in Computer Science and Engineering**
Location: Linköping University, Linköping, Sweden
‣ **Research Area:** LLM-based diagnosis and repair of Modelica models
‣ **Advisor:** [Dr. Daniel Varro](#)
‣ **Activities and Societies:** Committee member at [Navitas.se](#), the largest student association focused on sustainable development.
- Sep 2019 Oct 2022** **Master of Science - MSc in Information Technology (Subdiscipline: Enterprise Architecture)**
Location: Shahid Beheshti University, Tehran, Iran (Ranked 741th university in the world by QS 2026)
‣ **GPA:** 19.46 / 20 (1st Rank)
‣ **Thesis Title:** Using empirical data to support technology selection in software architecture decision-making
‣ **Advisor:** [Dr. Sadeqh Aliakbary](#)
‣ **Activities and Societies:** I coordinated with the executive team and the Third National Conference on Enterprise Architecture workshop lecturers. ([Team Page in Conference Website](#))
- Sep 2012 Jun 2018** **Bachelor of Science - BSc in Computer Engineering (Subdiscipline: Software Engineering)**
Location: Sharif University of Technology, Tehran, Iran (Ranked 375th university in the world by QS 2026)
‣ **Activities and Societies:** Initiated the Free Software Users Group, contributed to Rayanesh magazine, and supported Sharif AI Challenge 2015 technical infrastructure.

PUBLICATIONS

- Sadrnezhaad, M., Hernández López, J.A., Mårtensson, T., Varró, D.: *Generative AI in Simulation-Based Test Environments for Large-Scale Cyber-Physical Systems: An Industrial Study*. In: *Product-Focused Software Process Improvement (PROFES 2025)*, Salerno, Italy, December 1–3, 2025. Lecture Notes in Computer Science, vol. 16361, pp. 1–17. Springer, (2026).
[10.1007/978-3-032-12089-2_13](#) (Full research paper; CORE B; 37% acceptance rate.)

TEACHING EXPERIENCE

- **Accredited University Courses:**
‣ I contributed as a lab assistant, supporting course delivery through course planning, grading exams and assignments, assisting students in problem-solving sessions, and managing online course forums.

University	Course Title	Level	Examiner	Semester(s)
Linköping University	Methods and Tools for Large Distributed Projects	MSc	Dr. August Ernstsson	Winter 2026 , Winter 2025
	Concurrent Programming and Operating Systems	BSc		Winter 2026 , Winter 2025
Shahid Beheshti University	Advanced Software Engineering	MSc	Dr. Fereidoon Shams Aliee	Fall 2021
Sharif University of Technology	System Analysis and Design	BSc	Seyed Ebrahim Abtahi	Fall 2017 , Spring 2017 , Fall 2016
			Dr. Mehraneh Shantiaei	Spring 2017
	Advanced Programming in Java	BSc	Dr. Salman Marvasti	Fall 2015
	Technical Presentation	BSc	Dr. Ali M. A. Hemmatyar	Fall 2014

- **Thesis Supervision:**
‣ **Amin, M. and Rosengren, T.** *AI Agent for Asynchronous Interaction in Telecom Configuration Tool*. Master's thesis, Linköping University, Department of Computer and Information Science, Datateknik, 30 ECTS, 2026. (scheduled) **Supervisors:** M. Sadrnezhaad (Internal), I. Persson (External). **Examiner:** D. Varró. **External Commissioner:** Ericsson AB.
‣ **Paulsson, M.** *Improving Quality in AI-Generated Code through Prompt Engineering – An Evaluation of the Impact of Prompting Techniques on Software Quality in Code Generated by Large Language Models*. Master's thesis, Linköping University, Department of Computer and Information Science, Datateknik, 30 ECTS, 2025. [Available online](#). **Supervisors:** M. Sadrnezhaad (Internal), T. Mårtensson (External). **Examiner:** D. Varró. **External Commissioner:** SAAB AB.

External Extracurricular Courses:

- Taught 200+ students with high satisfaction rates: 99% believe the instructor knows the subject well, 98% find the course well-organized, 97% enjoy the current content and respectful, friendly style, and 91% appreciate prepared class plans.
- Developed courseware that includes announcements, resources, topics, calendars, assignments, and code samples.

Course Title	Organizer	Semesters
Python Programming	Sharif University of Technology - Sharif Rayan Academy	Spring 2019, Winter 2020
	University of Tehran - Cultural Education and Development Centre	Fall 2018
	Saipa Training Center	Summer 2019
	MapsHR Company	Fall 2021
	Shahid Beheshti University - Culture Home Center	Summer 2017, Fall 2017, Winter 2018, Summer 2018, Fall 2018, Winter 2019, Summer 2019
Front-end Web Development		Summer 2016, Fall 2018, Winter 2019, Summer 2019
Linux System Administration		Fall 2016
Web Scraping in Python		Winter 2018

Public Engagement & Invited Talks

- Dec 2020 **Animating and Simulating Business Processes to Redesign Them**, Informatics Society of Iran
- Apr 2020 **Semantic Storage and Retrieval Using the Resource Description Framework**, Shahid Beheshti University
- Dec 2019 **Simulating Self-Adaptive Software Architecture with the SWIM Scenario in the Rainbow Framework**, Faculty of Computer Science and Engineering, Shahid Beheshti University
- Dec 2017 **What is Net Neutrality, and Where Does it Come From?**, Sharif University of Technology
- Sep 2017 **A Comparative Introduction to MV* Patterns and Pattern Families**, Tehran Linux Users Group, Univ. of Tehran
- May 2016 **Mezzanine Content Management System Workshop (Pycon)**, Iran University of Science and Technology

HONORS & AWARDS

- May 2025 **Reviewer**, 41st International Conference on Logic Programming (ICLP 2025). [10.1017/S147106842510032X](https://doi.org/10.1017/S147106842510032X)
- Jul 2019 **Ranked 5th**, Nationwide Masters University Admission Exam among more than 15000 participants.
- Sep 2014 **23rd Team**, Sharif 5th Cyber Security and Hacking Competition (CTF) among over 800 teams. [Finals](#), [Quals](#)
- Nov 2013 **16th Team**, Sharif 4th Cyber Security and Hacking Competition (CTF) among over 100 teams. [Finals](#), [Quals](#)

WORK EXPERIENCE

- Nov 2022 - Sep 2024** **Technical Lead & DevOps**, [HUMMERS LEASING AND](#) [HUMMERS BROKER](#), Tehran, Iran
 - Formed a high-performing technical team after systemically evaluating 500+ individuals in various areas.
 - Maintained communication with stakeholders to thoroughly grasp project requirements, progress, and risks.
 - Implemented automation scripts for DevOps processes, including email server and GitLab administration.
 - Implemented enterprise-level web software for leasing management using Django and Bootstrap.

Team Leadership SDLC Nginx Redis Linux ATS Recruitment Django Zimbra Mail Server HTMX PostgreSQL GitLab CoCalc Moodle Selenium Agile Coaching
- Jul 2020 - Oct 2022** **Graduate Researcher, DATA-ORIENTED RESEARCH FOR SOFTWARE ANALYTICS LAB**, Tehran, Iran
 - Developed two recommender systems to help library selection. One used recent deep learning algorithms, while the other used content-based recommenders to alleviate the cold-start problem.
 - Extracted GitHub data (< 3 mins) using BigQuery, filtered incorrect libraries based on Maven conventions, and built samples using repository info (stars, branches, authors, commits, lifetimes, etc.).
 - Extracted project topics and descriptions from GitHub API and processed them to complete tags.
 - Designed an interactive dashboard using Data Studio for seamless data exploration.

Recommender System Deep Learning Python Google Cloud Google Colab BigQuery Data Studio Visual Paradigm
- Jul 2017 - Sep 2024** **Technical Lead & Web Developer**, [CONNECT ISTI](#), Tehran, Iran
 - Developed web software using Django, implemented black box regression tests, administered Linux servers, and established a continuous delivery pipeline and data backup strategies.
 - Created detailed technical documents for various stakeholders and documented recruitment procedures.
 - Collected requirements, analyzed them, and divided them into actionable implementation tasks.

Kanban System Analysis Software Architectural Design Django Semantic-UI PyCharm HTML5 CSS3 JavaScript
- Sep 2016 - Jul 2018** **Undergraduate Research Assistant, INTELLIGENT SOFTWARE ENGINEERING LAB**, Tehran, Iran
 - Developed a JetBrains plugin to track user actions and a tool for detecting potential bugs by analyzing GitHub histories to find commits later fixed in subsequent revisions; acknowledged in "Cross-project code clones in GitHub" ([10.1007/s10664-018-9648-z](https://doi.org/10.1007/s10664-018-9648-z)).

GitHub Datasets Java Standard pygit2 gensim nlTK GHTorrent code clone detection JGit JetBrains Plugin SQL